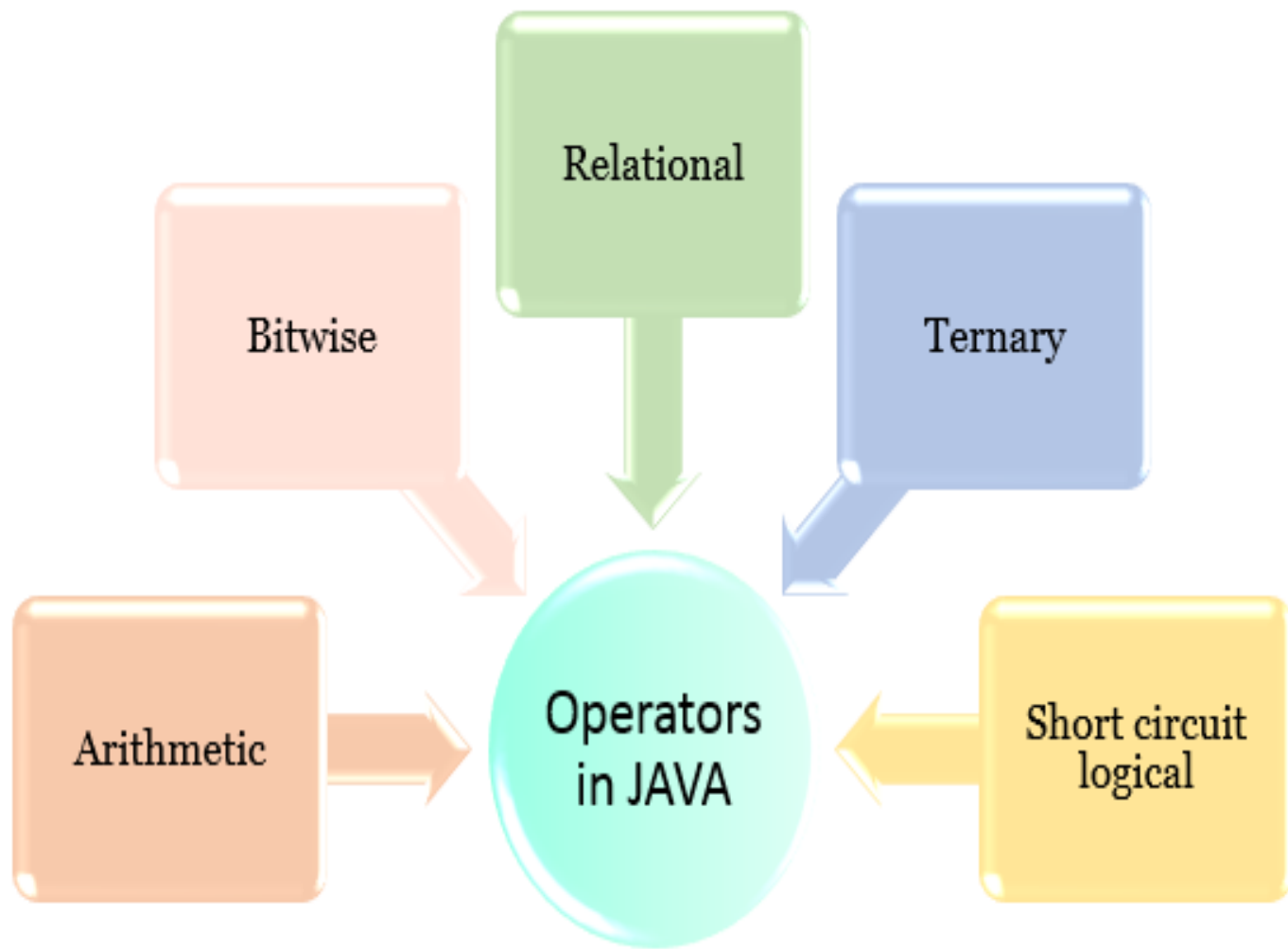




Arithmetic Operators

- addition: $+$
- subtraction: $-$
- multiplication: $*$
- division: $/$
- modulus: $\%$
- increment: $++$
- decrement: $--$

Arithmetic Operator Examples

- $9 + 4$ yields 13
- $9 - 4$ yields 5
- $9 * 4$ yields 36
- $9 / 4$ yields 2 (when two integers are divided, the remainder is thrown away)
- $9.0/4$ yields 2.25
- $9 \% 4$ yields 1 (the remainder when 9 is divided by 4)
- $x++$ and $++x$ increment the value of x by 1
- $x--$ and $--x$ decrement the value of x by 1

Relational Operators

Operator	Meaning
==	equal to
!=	not equal to
>	greater than
<	less than
>=	greater than or equal to
<=	less than or equal to

Assignment Operator

- General form:

var = expression;

- Variations:

`x = y = z = 3;`

`x += 10; // adds 10 to x`

- More compound assignment operators:

<code>+=</code>	<code>-=</code>	<code>*=</code>	<code>/=</code>
<code>%=</code>	<code>&=</code>	<code>!=</code>	<code>^=</code>

Logical Operators

Operator	Meaning
&	AND
	OR
^	XOR (exclusive OR)
	short-circuit OR
&&	short-circuit AND
!	NOT

This is a list of arithmetic operators in Java.

If a=10 and b=5, the result of each operation is mentioned in the Result column.

Operator	Operation	Expression	Result
+	Addition	a + b	15
-	Subtraction	a - b	5
*	Multiplication	a * b	50
/	Division	a / b	2
%	Modulus	a % b	0
--	Decrement	a--, --a	9
++	Increment	a++, ++a	11
+=	Addition assignment	a += b (same as a = a + b)	15
-=	Subtraction assignment	a -= b (same as a = a - b)	5
*=	Multiplication assignment	a *= b (same as a = a * b)	50
/=	Division assignment	a /= b (same as a = a / b)	2
%=	Modulus assignment	a %= b (same as a = a % b)	0

This is a list of Bitwise operators in Java. Here a=0 and b=1

Operator	Operation	Expression	Result
&	Bitwise AND	a & b	0
	Bitwise OR	a b	1
^	Bitwise exclusive OR	a ^ b	1
<<	Shift left	a << b	
>>	Shift right	a >> b	
~	Bitwise unary NOT	~a	1
&=	Bitwise AND assignment	a &= b is same as a = a & b	a=0
=	Bitwise OR assignment	a = b is same as a = a b	a=1
^=	Bitwise exclusive OR assignment	a ^= b is same as a = a ^ b	a=1
>>=	Shift right assignment	a >>= b is same as a = a >> b	
<<=	Shift left assignment	a <<= b is same as a = a << b	

Truth Table

p	q	p & q	p q	p ^ q	! p
False	False	False	False	False	True
True	False	False	True	True	False
False	True	False	True	True	True
True	True	True	True	False	False

Operator Precedence

High	++ (postfix)	-- (postfix)					
	++ (prefix)	-- (prefix)	~	!	+ (unary)	- (unary)	(type-cast)
	*	/	%				
	+	-					
	>>	>>>	<<				
	>	>=	<	<=	instanceof		
	==	!=					
	&						
	^						
	&&						
	? :						
	=	op=					
Low							

Automatic Type Conversions

- A variable of one type A can be assigned a value of another type B if:
 - the two types are compatible
 - the type A is larger than the type B
- In that case, the value is automatically converted from type B to type A.
- Example:

```
double x = 3;    // the integer 3 is converted to  
                 // the double 3.0
```

Type Conversion

- A variable of type A can be assigned a value of type B even if it isn't a widening conversion if it is compatible and you use a type cast.
- Example:

```
int x = (int) 3.14; // assigns 3 to x  
byte y = (byte) 256; // assigns 0 to b
```

```
class Demo{  
  
    public static void main(String[]  
args) {  
        int x=100;  
        int y=x;  
        x++;  
        int z=y;  
        z++;  
        System.out.println(y);  
    }  
}
```

What happens, when the program
executes?

101
102
0
100

Consider the following code inside Demo.java file

```
class Demo{  
  
    public static void main(String[] args){  
        int m = 1;  
        int n = ++m + m++ + --m;  
        System.out.println(n);  
    }  
}
```

What happens, when the program executes?

7
6
5
8

What will be the output of the following code?

```
public class LogicalDemo {  
  
    public static void main(String[] args) {  
        boolean x = true;  
        boolean y = false;  
  
        System.out.println(!(x && y));  
        System.out.println(x&&y);  
        System.out.println(x||y);  
  
    }  
}
```

true false true
true true false
false false true
true true false